Research Methods

Ground to Mastery — Study Notes

Research process, design, data collection, analysis, ethics

BSM 121 | Year 1 Semester 2 | DCS

Coursework 40% (Test 10% + Assignment 10% + Practical 20%) | Final Exam 60%

How to Study This Course

Research Methods is a systematic, conceptual course. Study by understanding how research progresses step-by-step. For the exam, know definitions precisely, know the differences between types, and be able to apply concepts to scenarios.

> 📝 *Assessment: Test 10% + Assignment 10% + Practical/Case Study 20% + Exam 60%. Practical = writing a short research proposal.*

UNIT 1: INTRODUCTION TO RESEARCH

Definition, concepts, language of research, qualities of a researcher

▸ 1.1 — What is Research?

Research is a systematic, controlled, empirical, and critical investigation of natural phenomena guided by theory and hypotheses about the presumed relationship between such phenomena.

Term / Concept	Definition / Explanation
`Research`	Systematic investigation to discover new knowledge, verify existing knowledge, or solve a problem
`Scientific research`	Research following scientific method — objective, testable, reproducible
`Empirical`	Based on observation and experiment — not just theory or opinion
`Systematic`	Follows organised, structured procedures — not random or informal
`Objective`	Free from personal bias — findings based on data, not feelings
`Replicable`	Other researchers can repeat the study and get similar results

▸ 1.2 — Language of Research (Key Terms)

Term / Concept	Definition / Explanation
`Variable`	A characteristic that can take different values — age, income, test score
`Dependent variable`	The outcome being measured — what changes as a result
`Independent variable`	The variable being manipulated or studied — the cause
`Hypothesis`	A testable statement predicting the relationship between variables
`Theory`	A well-supported explanation of a phenomenon based on evidence
`Concept`	An abstract idea or general idea — e.g. intelligence, poverty, crime
`Construct`	An abstract concept measured indirectly — e.g. motivation, satisfaction

`Operationalisation`	Defining how a variable will be measured — turning concept into measurable
`Population`	The entire group the researcher wants to study
`Sample`	A subset of the population selected for the study
`Data`	Raw facts collected during research
`Validity`	Does the study measure what it claims to measure?
`Reliability`	Would the study produce the same results if repeated?

▸ 1.3 — Qualities of a Good Researcher

- Curiosity — always asking questions and seeking answers
- Objectivity — no personal bias in collecting or interpreting data
- Patience — research takes time and repeated attempts
- Critical thinking — questioning assumptions and evaluating evidence
- Honesty and integrity — reporting findings truthfully, even unwanted results
- Organisation — systematic approach to data collection and record-keeping
- Communication skills — writing clearly and presenting findings effectively

UNIT 2: TYPES OF RESEARCH

Classification, survey, case study, feasibility, pilot study

▸ 2.1 — Classification of Research

Term / Concept	Definition / Explanation
`Basic/Pure research`	Generates new knowledge for its own sake — no immediate application — academic
`Applied research`	Solves specific practical problems — immediate real-world use
`Descriptive research`	Describes what exists — surveys, observations — no manipulation
`Analytical research`	Analyses existing data to understand relationships
`Quantitative research`	Uses numbers and statistics — surveys with Likert scales, experiments
`Qualitative research`	Uses words, observations — interviews, focus groups, case studies
`Mixed methods`	Combines quantitative and qualitative — most comprehensive
`Longitudinal`	Studies same subjects over long time period
`Cross-sectional`	Studies different groups at one point in time

▸ 2.2 — Related Forms of Research

Term / Concept	Definition / Explanation
`Survey research`	Collects data from large number of subjects using questionnaires or interviews
`Case study`	In-depth investigation of a single case — person, organisation, event
`Feasibility study`	Assesses whether a proposed project is practical and worthwhile
`Pilot study`	Small-scale preliminary study to test research instruments before main study
`Experimental research`	Researcher manipulates independent variable — controls conditions — laboratory
`Action research`	Researcher is also a participant — aims to solve immediate problem

UNIT 3: RESEARCH PROCESS

Steps from problem identification to report writing

▸ 3.1 — Research Process Steps

1. Identify and define the research problem
2. Review existing literature on the topic
3. Formulate research hypothesis and objectives
4. Select research design and methodology
5. Design data collection instruments (questionnaire, interview guide)
6. Collect data from sample
7. Process and analyse data (statistics, coding)
8. Interpret and discuss findings
9. Write research report and present conclusions
10. Publish and apply findings

> 📝 *Exam: Know all 10 steps of the research process in ORDER. You may be asked to arrange them or describe each step.*

UNIT 4: FORMULATING THE RESEARCH PROBLEM

Problem identification, objectives, hypothesis, research questions

▸ 4.1 — Problem Identification

Term / Concept	Definition / Explanation
`Research problem`	A gap in knowledge or an unsolved issue that research will address
`Sources of problems`	Personal experience, literature review, social issues, theory,

	supervisor suggestion
`Good research topic`	Feasible, interesting, novel, ethical, relevant — FINER criteria
`Problem statement`	A clear, concise description of the problem — what, where, who, when

▸ 4.2 — Research Objectives

Research objectives are specific statements of what the study aims to achieve. They should be SMART: Specific, Measurable, Achievable, Relevant, Time-bound.

Example: 'To determine the effect of mobile phone use on academic performance of Year 1 DCS students at [institution] during 2024.'

▸ 4.3 — Research Hypothesis

Term / Concept	Definition / Explanation
`Null hypothesis (H0)`	States there is NO relationship between variables: 'Phone use does not affect performance'
`Alternative hypothesis (H1)`	States there IS a relationship: 'Phone use affects academic performance'
`Directional hypothesis`	Specifies the direction: 'More phone use leads to lower performance'
`Non-directional hypothesis`	States a relationship exists without specifying direction

UNIT 5: LITERATURE REVIEW

Purpose, scope, steps, sources, databases

▸ 5.1 — What is a Literature Review?

A literature review is a critical summary and analysis of existing research on your topic. It shows what is already known, identifies gaps, and justifies your study.

▸ 5.2 — Purposes of Literature Review

- Understand what is already known about the topic
- Identify gaps that your research will fill
- Provide theoretical framework for your study
- Avoid duplicating existing research
- Help refine research questions and methodology

▸ 5.3 — Sources of Literature

Term / Concept	Definition / Explanation
`Primary sources`	Original research — journal articles, theses, conference papers, original data
`Secondary sources`	Summaries of primary — textbooks, review articles, encyclopaedias
`Tertiary sources`	Indexes and abstracts — databases, bibliographies — guide you to primary/secondary
`Online databases`	Google Scholar, PubMed, JSTOR, ResearchGate, IEEE Xplore, ERIC
`Grey literature`	Government reports, NGO publications, working papers — not peer-reviewed

▸ 5.4 — Steps in Literature Review

1. Define your topic and keywords
2. Search databases using keywords and Boolean operators (AND, OR, NOT)
3. Select relevant, recent, credible sources
4. Read critically — note key arguments, methods, findings, gaps
5. Organise by themes or chronologically
6. Write — synthesise, don't just summarise one source after another
7. Cite all sources properly (APA format)

UNIT 6: RESEARCH DESIGN & DATA COLLECTION

Design types, qualitative and quantitative methods, questionnaire design

▸ 6.1 — Research Design

Term / Concept	Definition / Explanation
`Experimental`	Researcher controls variables — treatment and control groups — lab or field
`Quasi-experimental`	Like experimental but no random assignment — real-world settings
`Descriptive`	Describes a phenomenon as it is — surveys, observation
`Correlational`	Looks for relationships between variables — no manipulation
`Case study`	In-depth study of single case — multiple data sources
`Ethnographic`	Researcher immerses in culture/community — long-term observation

▸ 6.2 — Quantitative Data Collection

Term / Concept	Definition / Explanation
`Questionnaire`	Written set of questions — can reach many people — structured responses
`Structured interview`	Interviewer asks same questions to all participants — formal
`Observation (structured)`	Observing and recording using pre-defined categories
`Experiments`	Controlled tests measuring effect of manipulated variable
`Secondary data`	Existing data — government records, databases, published statistics

▸ 6.3 — Qualitative Data Collection

Term / Concept	Definition / Explanation
`Unstructured interview`	Open conversation — flexible — explore participant's views in depth
`Semi-structured interview`	Mix of set questions and open exploration — most common
`Focus Group Discussion (FGD)`	Group of 6-12 people discussing a topic — reveals social views
`Observation (unstructured)`	Researcher observes without checklist — notes everything relevant
`Document analysis`	Analysing existing texts, photos, videos, records

▸ 6.4 — Questionnaire Design Principles

- Clear and simple language — avoid jargon and double negatives
- One idea per question — avoid double-barrelled questions
- Avoid leading questions that suggest the desired answer
- Logical flow — general to specific, easy to hard
- Mix of question types — open, closed, Likert scale
- Pilot test with small group before main study
- Include introduction and consent section at start

▸ 6.5 — Types of Questions

Term / Concept	Definition / Explanation
`Open-ended`	No fixed answers — 'What do you think about...?' — rich qualitative data
`Closed-ended`	Fixed options — 'Yes/No', multiple choice — easy to quantify
`Likert scale`	1-5 or 1-7 agreement scale — Strongly Agree to Strongly Disagree
`Rating scale`	Rate from 1-10 — satisfaction, frequency

Dichotomous	Two options only — Yes/No — Male/Female
Ranking	Rank options in order of preference

UNIT 7: SAMPLING TECHNIQUES

Probability and non-probability sampling, sample size

▸ 7.1 — Key Sampling Concepts

Term / Concept	Definition / Explanation
Population	All members of the group the study is about
Sample	The subset of population selected for the study
Sampling frame	Complete list of all members of the population — used to select sample
Representativeness	Sample should reflect characteristics of the whole population
Sampling error	Difference between sample results and true population values
Bias	Systematic error in how sample was selected — makes it unrepresentative

▸ 7.2 — Probability Sampling (Random)

Term / Concept	Definition / Explanation
Simple random sampling	Each member has equal chance — lottery method, random number table
Systematic sampling	Select every kth member — k = population size / sample size
Stratified sampling	Divide population into strata, then random sample from each — e.g. by year/gender
Cluster sampling	Randomly select groups (clusters), then include all or sample from clusters
Multi-stage sampling	Combination of methods — clusters then random within clusters

▸ 7.3 — Non-Probability Sampling

Term / Concept	Definition / Explanation
Purposive/Judgement	Researcher deliberately selects specific subjects — experts, key informants
Convenience/Accidental	Select whoever is available and willing — easiest but least representative
Snowball	Participants recruit others — used for hard-to-reach populations
Quota	Set quotas for categories (e.g. 50 male, 50 female) — then convenience within quota

▸ 7.4 — Sample Size Determination

Sample size depends on: population size, desired confidence level (usually 95%), acceptable margin of error (usually 5%), expected variability in the data, and available resources.

Yamane's formula: $n = N / (1 + N(e^2))$ where N = population size, e = margin of error (0.05 for 5%)

UNIT 8: DATA ANALYSIS, ETHICS & REPORT WRITING

Processing data, qualitative and quantitative analysis, plagiarism, APA referencing

▸ 8.1 — Data Processing

Term / Concept	Definition / Explanation
`Editing`	Check completed questionnaires for completeness, consistency, legibility
`Coding`	Assign numerical codes to responses — allows computer analysis
`Cleaning`	Check for errors, outliers, missing data — verify and correct
`Entry`	Enter coded data into spreadsheet or SPSS — double-entry to verify accuracy

▸ 8.2 — Quantitative Analysis

Term / Concept	Definition / Explanation
`Descriptive statistics`	Summarise data — mean, median, mode, range, standard deviation, frequency, percentage
`Mean`	Average: sum of values / number of values
`Median`	Middle value when arranged in order — not affected by outliers
`Mode`	Most frequently occurring value
`Standard deviation`	Measure of spread — how far values are from the mean
`Correlation`	Measures strength and direction of relationship — Pearson r (-1 to +1)
`Cross-tabulation`	Compare two variables — shows relationship between categories

▸ 8.3 — Qualitative Analysis

Term / Concept	Definition / Explanation
`Coding`	Label segments of text with themes — manual or software (NVivo, Atlas.ti)
`Thematic analysis`	Identify, analyse, and report patterns (themes) across data

`Content analysis`	Systematically categorise and count content — bridges qualitative and quantitative
`Narrative analysis`	Analyse stories people tell about their experiences
`Discourse analysis`	Analyse how language is used and what it reveals about power and society

▸ 8.4 — Ethical Issues in Research

Term / Concept	Definition / Explanation
`Informed consent`	Participants must be told the purpose, risks, and voluntarily agree
`Confidentiality`	Data kept private — names not revealed unless participant agrees
`Anonymity`	Researcher cannot link responses to specific individuals
`No harm principle`	Research must not physically, psychologically, or socially harm participants
`Right to withdraw`	Participants can leave the study at any time without penalty
`Honesty`	Do not fabricate, falsify or selectively report data
`Plagiarism`	Using others' work without credit — see below

▸ 8.5 — Plagiarism

Term / Concept	Definition / Explanation
`Copy and paste`	Copying text from a source without quotation and citation
`Paraphrasing`	Rewording someone's ideas without citing the source
`Self-plagiarism`	Reusing your own previously submitted work without disclosure
`Ghost writing`	Submitting work written by someone else as your own
`Fabrication`	Making up data or citations
`Prevention`	Cite all sources, use plagiarism checkers (Turnitin), quote and paraphrase properly

▸ 8.6 — APA Referencing (7th Edition)

APA 7th Edition format

```
// JOURNAL ARTICLE
Author, A. A., & Author, B. B. (Year). Title of article.
    Journal Name, Volume(Issue), Page–Page.
    https://doi.org/xxxxx

// Example:
Creswell, J. W., & Creswell, J. D. (2018). Research design:
    Qualitative, quantitative, and mixed methods approaches (5th ed.).
```

```
    SAGE Publications.

// BOOK
Author, A. A. (Year). Title of book. Publisher.

// WEBSITE
Author, A. A. (Year, Month Day). Title of webpage.
    Website Name. URL

// In-text citation:
(Creswell & Creswell, 2018)
Creswell and Creswell (2018) argued that...
```

> 📝 *EXAM: Know the difference between primary, secondary, and tertiary sources. Know the difference between probability and non-probability sampling. Know all steps of the research process. Know what plagiarism is and how to prevent it.*